

1 Supplementary Information: How temperature, photoperiod,
2 and evolutionary history shape forest leafout

3

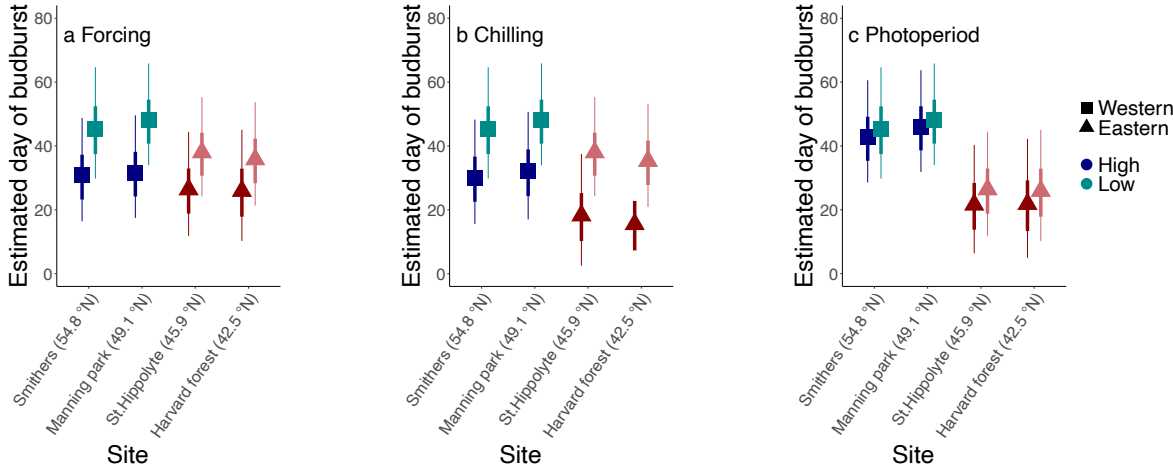


Figure S1: Estimated day of budburst in response to forcing (a) across sites under low chilling and short photoperiods, chilling across sites under low forcing and short photoperiods (b), and across photoperiods (c) under low forcing and chilling for species sampled from our four sites. The thin bars represent the 90% uncertainty interval, while the thicker bars represent the 50% uncertainty interval, and symbols the mean.

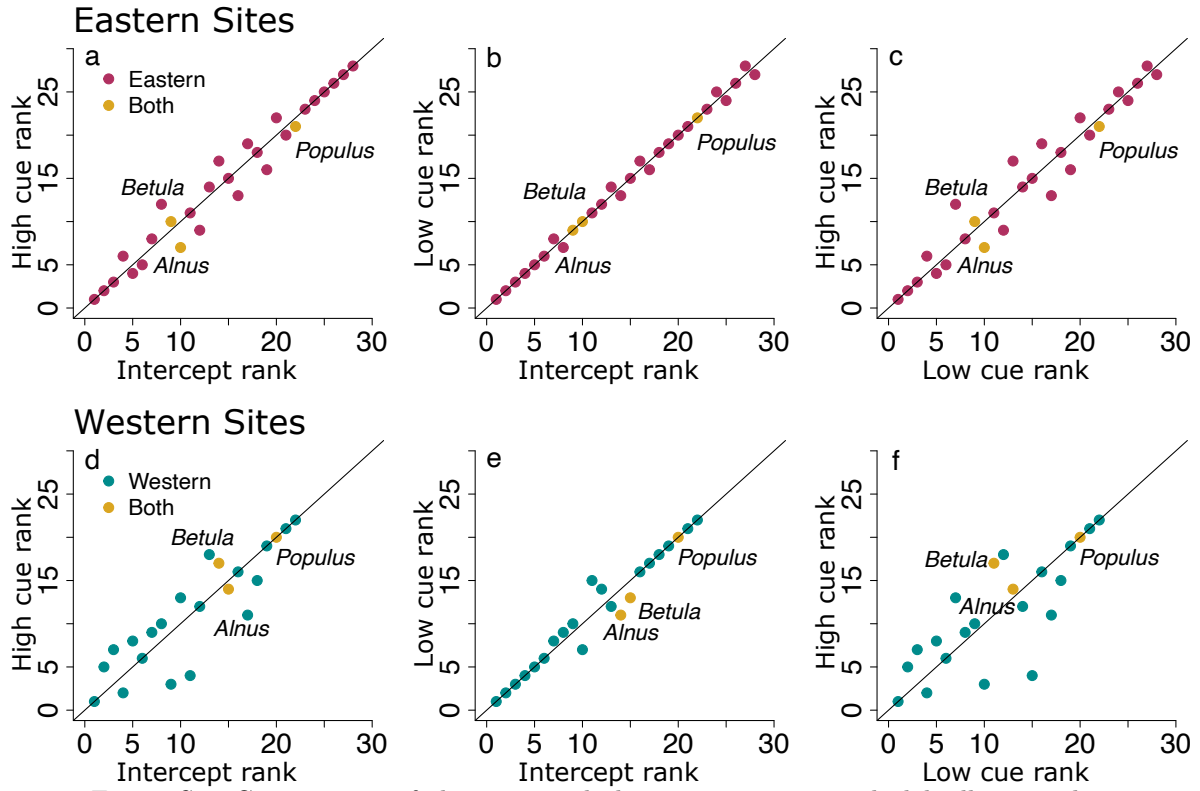


Figure S2: Comparisons of the estimated changes in species ranked budburst order, between species-level effects (species intercept) and under high cues (**a, d**), species-level effects and under low cues (**b, e**), and under low versus high cues (**c, f**), for our eastern species (in red, **a-c**) and western species (in blue, **d-f**). For the three species that occur in both transect (shown in yellow), *Alnus incana* exhibited the greatest rank change with a difference of three, while *Betula papyrifera* had a rank difference of two and *Populus tremuloides* a rank difference of one.

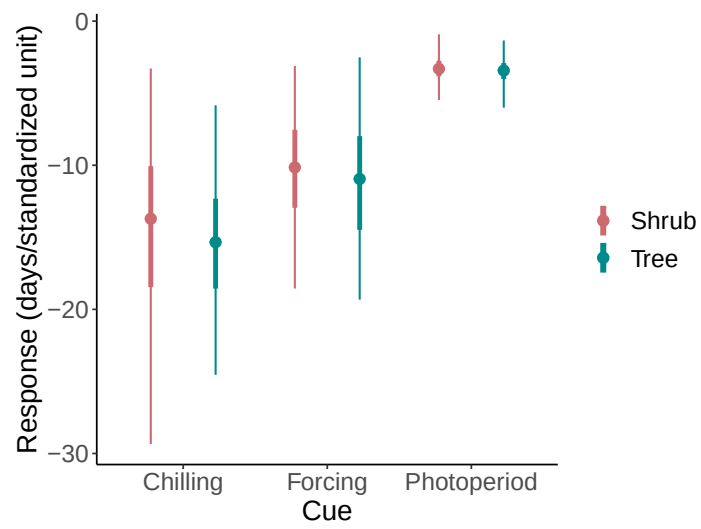


Figure S3: Comparisons of posterior distributions for cue estimates between shrub and tree species. Circles represent the mean, while the thicker error bars represent the 50% uncertainty interval, and thinner line the 90% quantile interval.

Table S1: Mean day of budburst following the start of forcing treatments, estimated from raw observational data of 47 species at our two western sites, E.C. Manning park and Smithers B.C., Canada, and our two eastern sites, Harvard Forest, MA, USA and St. Hippolyte, QC, Canada.

Species	Spp. abbreviation	Harvard forest	St. Hippolyte	Manning park	Smithers
<i>Acer glabrum</i>	acegla			36	39
<i>Acer pensylvanicum</i>	acepen	16	18		
<i>Acer rubrum</i>	acerub	22	25		
<i>Acer saccharum</i>	acesac	45	36		
<i>Alnus incana</i>	alninc			28	30
<i>Alnus incana</i>	alninc	33	25		
<i>Alnus viridis</i>	alnvir			44	43
<i>Amelanchier alnifolia</i>	amealn			19	18
<i>Aronia melanocarpa</i>	aromel	14			
<i>Betula alleghaniensis</i>	betall	20	21		
<i>Betula lenta</i>	betlen	30			
<i>Betula papyrifera</i>	betpap				31
<i>Betula papyrifera</i>	betpap	17	18		
<i>Corylus cornuta</i>	corcor	25	19		
<i>Cornus stolonifera</i>	corsto			15	17
<i>Fagus grandifolia</i>	faggra	42	43		
<i>Fraxinus nigra</i>	franig	38	38		
<i>Hamamelis virginiana</i>	hamvir	44			
<i>Ilex mucronata</i>	ilemuc	16	15		
<i>Kalmia angustifolia</i>	kalang	30	32		
<i>Lonicera canadensis</i>	loncan	17	16		
<i>Lonicera involucrata</i>	loninv			22	20
<i>Lyonia ligustrina</i>	lyolig	31			
<i>Menziesia ferruginea</i>	menfer			43	46
<i>Nyssa sylvatica</i>	nyssyl	32			
<i>Populus balsamifera</i>	popbal			30	31
<i>Populus grandidentata</i>	popgra	33	31		
<i>Populus tremuloides</i>	poptre			46	35
<i>Prunus pensylvanica</i>	prupen	18	16		
<i>Quercus alba</i>	quealb	45			
<i>Quercus rubra</i>	querub	36	34		
<i>Quercus velutina</i>	quevel	52			
<i>Rhamnus frangula</i>	rhafran	32			
<i>Rhododendron albiflorum</i>	rhoalb			19	
<i>Rhododendron prinophyllum</i>	rhopri	29			
<i>Ribes lacustre</i>	riblac			29	23
<i>Rubus parviflorus</i>	rubpar			28	30
<i>Sambucus racemosa</i>	samrac			33	
<i>Shepherdia canadensis</i>	shecan			25	24
<i>Sorbus scopulina</i>	sorsco			21	19
<i>Spiraea alba</i>	spialb	18	20		
<i>Spiraea betulifolia</i>	spibet			24	18
<i>Spiraea pyramidata</i>	spipyr			26	22
<i>Symphoricarpos albus</i>	symalb			27	32
<i>Vaccinium membranaceum</i>	vacmem			22	23
<i>Vaccinium myrtilloides</i>	vacmyr	13	17		
<i>Viburnum cassinoides</i>	vibcas	15	18		
<i>Viburnum edule</i>	vibedu			19	8
<i>Viburnum lantanoides</i>	viblan ⁴	31	28		

Table S2: Approximate chill units from our two western sites, E.C. Manning park and Smithers B.C., Canada, and our two eastern sites, Harvard Forest, MA, USA and St. Hippolyte, QC, Canada. Weather data for the fall of 2019 and winter 2020 was obtained from the Hope Slide weather station for our E.C. Manning park sites and the Smithers airport weather station for our Smithers site (Environment and Climate Change Canada, 2026). For our eastern sites, weather data for the fall of 2014 and winter 2015 was obtained from weather stations at Harvard Forest (Boose, E and Gould, E and VanScoy, M, 2026) and the St. Hippolyte weather station.(Environment and Climate Change Canada, 2026)

Site	Chilling treatment	Chill hours	Utah model	Chill portions
Harvard forest	Field chilling	892	814.50	56.62
Harvard forest	Field chilling + 30 d at 4°C	2140	2062.50	94.06
St. Hippolyte	Field chilling	682	599.50	44.63
St. Hippolyte	Field chilling + 30 d at 4°C	1930	1847.50	82.06
Smithers	Field chilling + 21 d at 4°C	1317	1368.00	54.95
Smithers	Field chilling + 56 d at 4°C	1965	2016.00	74.67
Manning park	Field chilling + 21 d at 4°C	1213	1377.00	55.09
Manning park	Field chilling + 56 d at 4°C	1861	2025.00	75.33

Table S3: Summary output from a phylogenetic Bayesian model in which species are partially pooled and phylogeny is included on the intercept. See the statistical analysis section of the methods for more detail.

	mean	5%	25%	75%	95%
α_{sp}	30.41	25.96	28.55	32.15	35.23
λ	0.43	0.08	0.27	0.59	0.75
$\beta_{forcing}$	-10.62	-11.90	-11.17	-10.09	-9.34
$\beta_{photoperiod}$	-3.36	-3.97	-3.61	-3.10	-2.73
$\beta_{chilling}$	-14.81	-16.69	-15.60	-14.03	-12.91
$\beta_{Manning\ park}$	2.09	1.45	1.82	2.35	2.74
$\beta_{Harvard\ forest}$	-6.62	-8.38	-7.30	-5.91	-4.93
$\beta_{St.\ Hippolyte}$	-8.76	-10.41	-9.43	-8.10	-7.19
$\beta_{forcing \times photoperiod}$	0.83	-0.46	0.31	1.35	2.13
$\beta_{forcing \times chilling}$	8.37	6.98	7.79	8.93	9.79
$\beta_{photoperiod \times chilling}$	-0.63	-2.00	-1.18	-0.07	0.75
$\beta_{photoperiod \times Manning\ park}$	0.55	-0.77	0.01	1.07	1.84
$\beta_{chilling \times Manning\ park}$	-0.35	-3.05	-1.37	0.74	2.13
$\beta_{forcing \times Harvard\ forest}$	3.79	1.73	2.94	4.65	5.87
$\beta_{photoperiod \times Harvard\ forest}$	-1.56	-2.98	-2.14	-0.97	-0.11
$\beta_{chilling \times Harvard\ forest}$	8.59	5.61	7.43	9.77	11.55
$\beta_{forcing \times St.\ Hippolyte}$	4.86	2.83	4.07	5.67	6.85
$\beta_{photoperiod \times St.\ Hippolyte}$	-2.15	-3.46	-2.70	-1.60	-0.82
$\beta_{chilling \times St.\ Hippolyte}$	7.18	4.65	6.18	8.22	9.68

Table S4: Summary output from a phylogenetic Bayesian model in which the phylogeny is included on the residuals. See the statistical analysis section of the methods for more detail.

	mean	5%	25%	75%	95%
α_{sp}	28.90	26.20	27.80	29.98	31.62
λ	0.49	0.05	0.25	0.75	0.96
$\beta_{forcing}$	-10.60	-11.90	-11.14	-10.12	-9.35
$\beta_{photoperiod}$	-3.30	-3.99	-3.60	-3.08	-2.72
$\beta_{chilling}$	-14.80	-16.72	-15.58	-14.04	-12.96
$\beta_{Manning\ park}$	2.10	1.45	1.84	2.34	2.73
$\beta_{Harvard\ forest}$	-5.90	-7.57	-6.58	-5.26	-4.30
$\beta_{St.\ Hippolyte}$	-8.10	-9.68	-8.75	-7.50	-6.59
$\beta_{forcing \times photoperiod}$	0.90	-0.42	0.32	1.39	2.12
$\beta_{forcing \times chilling}$	8.40	6.96	7.80	8.95	9.82
$\beta_{photoperiod \times chilling}$	-0.70	-2.15	-1.26	-0.09	0.78
$\beta_{forcing \times Manning\ park}$	-1.70	-3.11	-2.29	-1.20	-0.35
$\beta_{photoperiod \times Manning\ park}$	0.60	-0.75	0.05	1.11	1.84
$\beta_{chilling \times Manning\ park}$	-0.30	-2.94	-1.32	0.68	2.22
$\beta_{forcing \times Harvard\ forest}$	3.80	1.76	2.95	4.66	5.92
$\beta_{photoperiod \times Harvard\ forest}$	-1.60	-3.05	-2.15	-0.98	-0.14
$\beta_{chilling \times Harvard\ forest}$	8.80	5.89	7.62	10.01	11.79
$\beta_{forcing \times St.\ Hippolyte}$	4.90	2.86	4.03	5.67	6.88
$\beta_{photoperiod \times St.\ Hippolyte}$	-2.20	-3.53	-2.71	-1.60	-0.79
$\beta_{chilling \times St.\ Hippolyte}$	7.40	4.92	6.40	8.35	9.78

Table S5: Proportion of surviving samples per species for which budburst was observed in our controlled environment study of western sites. See (Flynn and Wolkovich, 2018) for survival in our study of eastern sites.

Species name	Proportion budburst	Plant type
<i>Acer glabrum</i>	0.83	tree
<i>Alnus incana</i>	1.00	shrub
<i>Alnus viridis</i>	0.92	shrub
<i>Amelanchier alnifolia</i>	0.99	shrub
<i>Betula papyrifera</i>	1.00	tree
<i>Cornus stolonifera</i>	0.99	shrub
<i>Lonicera involucrata</i>	0.87	shrub
<i>Menziesia ferruginea</i>	0.80	shrub
<i>Populus balsamifera</i>	0.98	tree
<i>Populus tremuloides</i>	0.90	tree
<i>Rhododendron albiflorum</i>	1.00	shrub
<i>Ribes lacustre</i>	0.82	shrub
<i>Rubus parviflorus</i>	0.94	shrub
<i>Sambucus racemosa</i>	0.95	shrub
<i>Shepherdia canadensis</i>	1.00	shrub
<i>Sorbus scopulina</i>	0.99	shrub
<i>Spiraea betulifolia</i>	0.94	shrub
<i>Spiraea pyramidata</i>	0.92	shrub
<i>Symphoricarpos albus</i>	0.84	shrub
<i>Vaccinium membranaceum</i>	0.90	shrub
<i>Viburnum edule</i>	1.00	shrub

4 Bibliography

- 5 Boose, E and Gould, E and VanScoy, M, 2026. Harvard Forest Climate Data since 1964. Harvard
6 Forest Data Archive: HF300 (v.14). Environmental Data Initiative. URL [https://doi.org/10.](https://doi.org/10.6073/pasta/ae08ef9f01d8c5509a31eff749cefb9)
7 [6073/pasta/ae08ef9f01d8c5509a31eff749cefb9](https://doi.org/10.6073/pasta/ae08ef9f01d8c5509a31eff749cefb9).
- 8 Environment and Climate Change Canada, 2026. Historical Data. URL [https://climate.weather.](https://climate.weather.gc.ca/historical_data/search_historic_data_e.html)
9 [gc.ca/historical_data/search_historic_data_e.html](https://climate.weather.gc.ca/historical_data/search_historic_data_e.html).
- 10 Flynn, D. F. B., and E. M. Wolkovich. 2018. Temperature and photoperiod drive spring phenology
11 across all species in a temperate forest community. *New Phytologist* **219**:1353–1362.